

Chapter 7

Verification and Validation of Simulation Models

Overview

7.1 Verification and validation

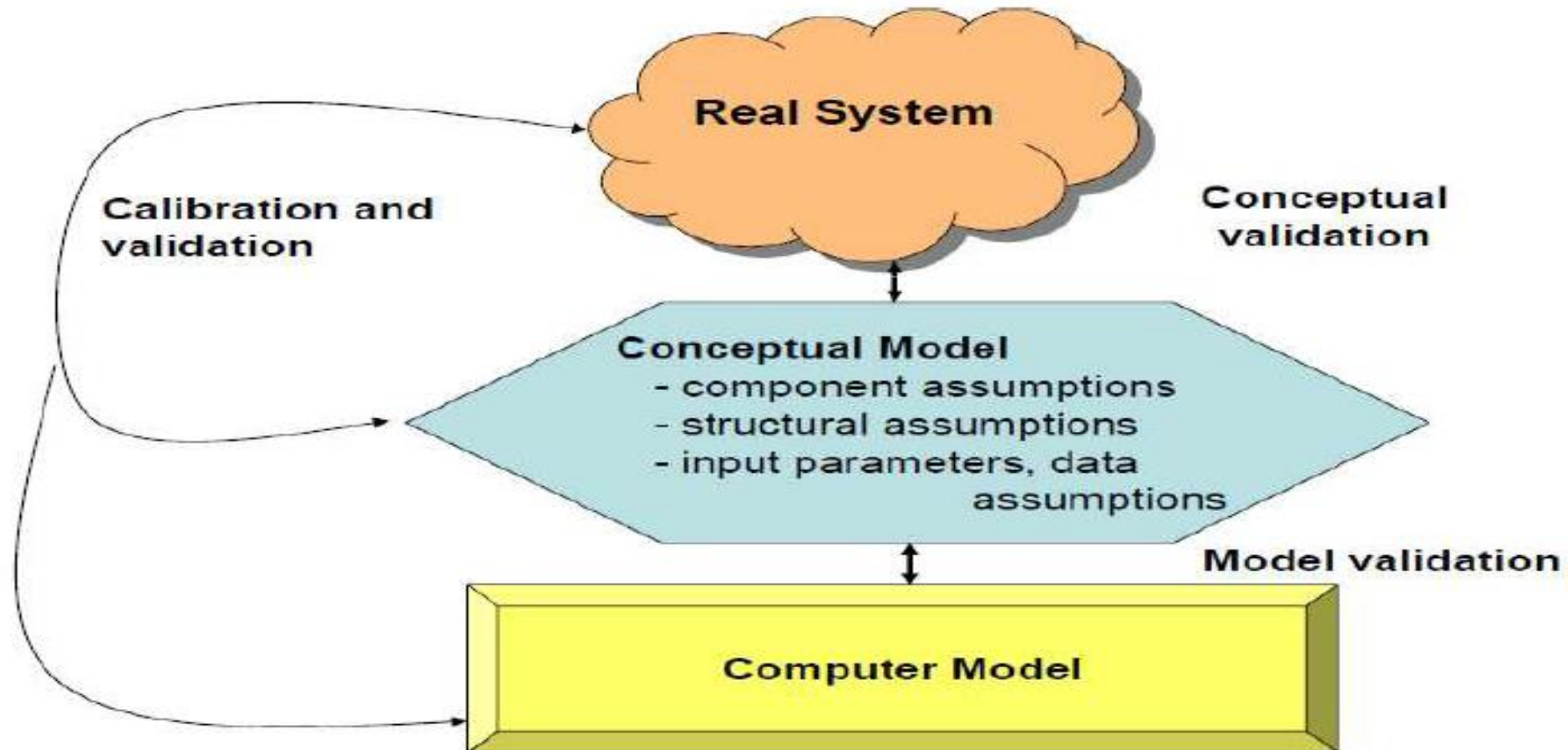
7.2 Verification of simulation models

7.3 Calibration and validation of models

7.4 Naylor and finger validation process

7.5 Validation: Errors

From modeling to simulation: model building, verification and validation



7.1 Verification and Validation

Verification

- Verification of a model is the process of confirming that it is correctly implemented with respect to the conceptual model (it matches specifications and assumptions deemed acceptable for the given purpose of application)
- Deals with building the model correctly, correctly implemented with good input and structure
- Verification pertains to the computer program prepared for the simulation model.
- Is the computer program performing properly?
- concerned with building the ***model right***.
- If the input parameters and logical structure of the model are correctly represented in the computer, verification has been completed.
- Verification answers the question "Have we built the model right?".

7.1 Verification and Validation

Validation

- concerned with building the *right model*.
- building the correct model, an accurate representation of the real system
- Validation is the determination that a model is an accurate representation of the real system.
- Validation is usually achieved through the calibration of the model, an iterative process of comparing the model to actual system behavior and using the discrepancies between the two, and the insights gained to improve the model.
- This process is repeated until model accuracy is judged acceptable.
- validation answers the question "Have we built the right model?".

7.2 Verification of Simulation Models

- Verification makes sure the conceptual model is translated into a computerized model properly and truthfully.
- Typically, three intuitive mechanisms can be used.

1. Common sense

- Have the computerized representation checked by someone other than its developer.
- Make a flow diagram which includes each logically possible action of a system can take when an event occurs, and follow the model logic for each action for each event type.
- Closely examine the model output under a variety of settings of input parameters.
- debugger is an essential component of successful simulation model building, which allows the analyst or the programmer to test the simulation program interactively.
- If one is modeling a certain queueing system, then simulation results can be compared with theoretical results as upper or lower bound.

7.2 Verification of Simulation Models

2. Through documentation:

- by describing in detail what the simulation programmer has done in design and implementing the program, one can often identify some problems.

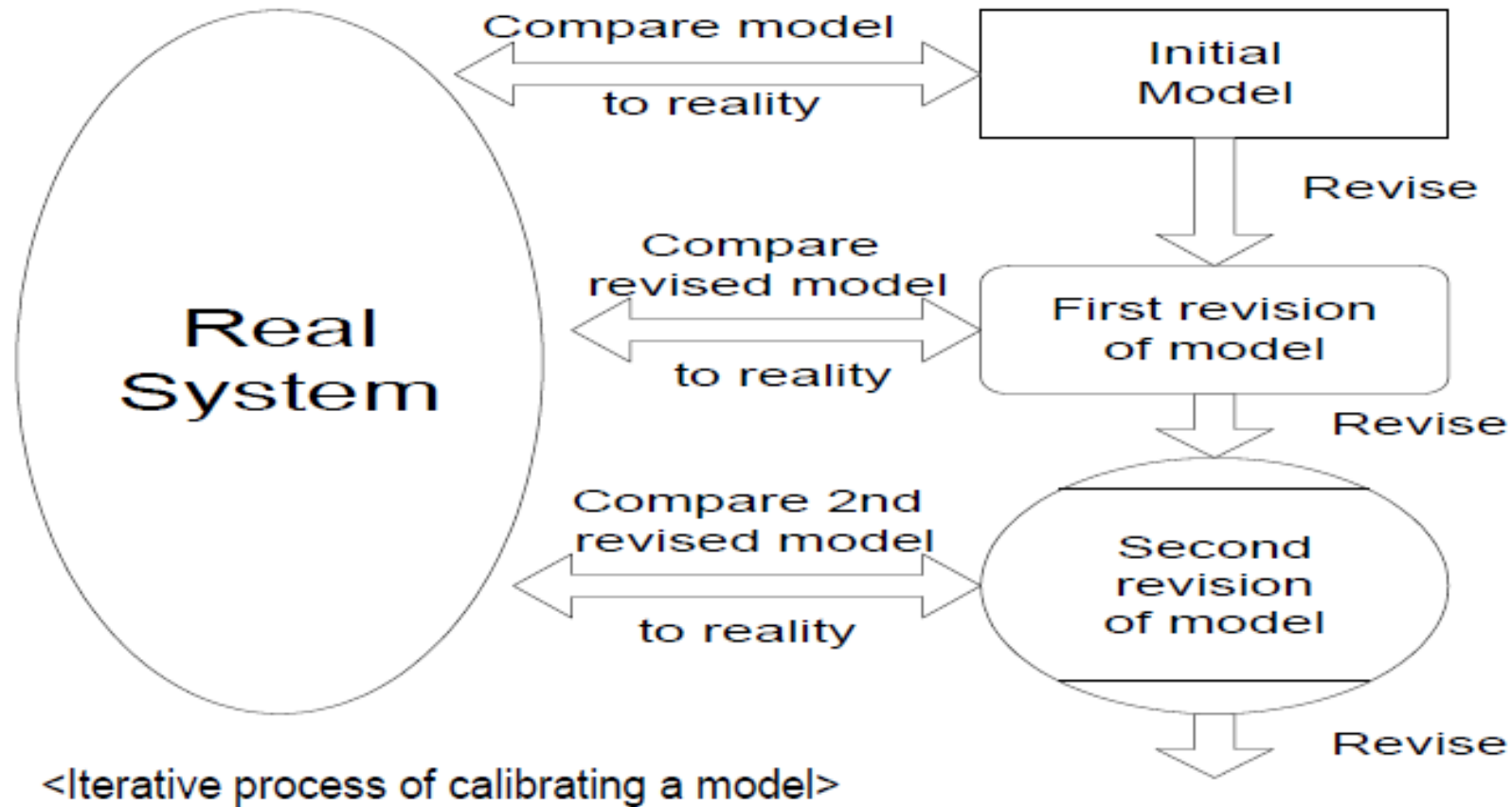
3. Trace:

- go through a few steps in an actual program to see if the behavior is reasonable

7.3 Calibration and Validation of Models

- Validation is the overall process of comparing the model and its behavior to the real system and its behavior.
- Calibration is the iterative process of comparing the model to the real system.
- Making adjustment or changes to the model, comparing the revised model to reality and so on.
- The comparison of the model to reality is carried out by subjective and objective tests.
 - A subjective test involves talking to people, who are knowledgeable about the system making models and forming the judgment.
 - Objective tests involve one or more statistical tests to compare some model output with the assumptions in the model
- Therefore, Calibration is the iterative process of comparing the model with real system, revising the model if necessary, comparing again, until a model is accepted (validated).

7.3 Calibration and Validation of Models



7.4 Naylor and Finger validation process

- Naylor and Finger formulated a three-step approach to the validation process
 1. Build a model that has high face validity.
 2. Validate model assumptions
 3. Compare the model input-output transformations to corresponding input-output transformation for the real system. (Validating input-output transformation)

7.3 Calibration and Validation of Models

1. Face Validity

- A model that has face validity appears to be reasonable imitation of real world system to people who are knowledgeable about real world system.
- Face validity is tested by having users and people knowledgeable with system, examine model output for reasonableness and in the process identify deficiencies.
- An added advantage of having the users involved in validation is that the model's credibility to users and user's confidence in model increases.

7.3 Calibration and Validation of Models

2. Validation of Model Assumptions

- Model assumptions fall into two categories: structural assumptions and data assumptions.
- **Structural assumptions** deal with such questions as how the system operates, how it is physically arranged, what kind of model should be used.
 - For example, number of servers in a fast food drive through line and if there is more than one, how are they utilized? Do the servers work in parallel where a customer completes a transaction by visiting a single server or does one server take orders and handle payment while other prepares and serves the order.
 - The system structure and operations should be verified with users of actual system.

7.3 Calibration and Validation of Models

2. Validation of Model Assumptions

- **Data assumptions:** what kind of input data model uses? What are the parameter values to the input data model?
- There must be sufficient amount of appropriate data available to build the conceptual model and validate the model.
- Data should be verified to come from a reliable source.
- Any outlier in data should be checked.

7.3 Calibration and Validation of Models

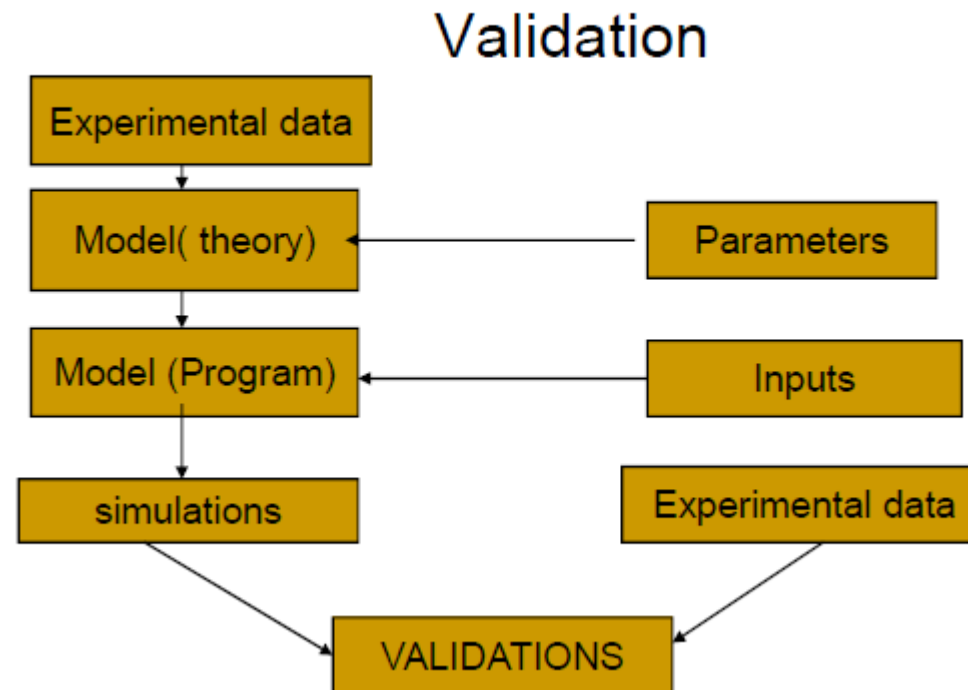
3. Validating Input-Output Transformations

Compare the model input-output transformations to corresponding input-output transformation for the real system.

- View the model as a black box
- Feed the input at one end and examine the output at the other end
- Use the same input for a real system, compare the output with the model output
- If they fit closely, the black box seems working fine
- Otherwise, something is wrong
- Data recorded while observing the system must be available in order to perform this test.

Model Validation

- Model validation is a necessary requirement for model application.
- To do a reliable validation, several steps must be taken and each of them may be a source of errors which will influence the final result.



7.5 Validation: Errors

- As a general rule, if there are discrepancies between observed and simulated data, the technical structure of a model should be the last factor to suspect.
 1. Model inadequate
 2. Lack of calibration
 3. Errors in the code
 4. Errors in the inputs
 5. Errors in the use
 6. Errors in the experimental data

Possible Solutions

1. Model Inadequacy

- Are all the important processes for a given environment included?
- Are the processes modeled correctly?
- Was the range of data used to develop model components for process simulation wide enough to include our conditions?

2. Lack of calibration

- Perform calibration till the model is in its accepted form

Possible Solutions

3. Errors in the Code

- Following steps can be undertaken to check a code:
 - Do calculations using for instance a spreadsheet and compare with model results.
 - Verify that simulation results are within the known physical and biological reality.
 - Run simulations with highly contrasting inputs.

4. Errors in the Experimental Data

- The experimental data used to test model predictive capabilities are affected by experimental error, which can be large.
- Only a large number of experimental data allows a meaningful evaluation of model performance in statistical terms.

References

- Jerry Banks, John S. Carson, II Barry L. Nelson , David M. Nicol,
“Discrete Event system simulation”
- <https://www.eg.bucknell.edu>